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LEARNER GUIDE

For

Certified Lean Six Sigma Yellow Belt (CLSSYB) Training

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Conducted by

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1	1 June 2026	Initial release — CLSSYB 2-day lesson plan.	Dr. Alfred Ang
2	15 June 2026	Refreshed labs against the CSSC Yellow Belt reference.	Dr. Alfred Ang
3	1 July 2026	Aligned the schedule with the Contoso Service Desk case-study scenario.	Dr. Alfred Ang
4	19 July 2026	Major content expansion: course restructured to follow the DMAIC roadmap across six topics with 14 labs.	Dr. Alfred Ang
5	19 July 2026	Converted to the non-funded commercial course standard; the course code is now C524. The 165 minutes previously reserved for formal evaluation on Day 2 are returned to teaching — elective labs 11-14 are now delivered in class. Rebuilt from a single source (course_data.py + data_domain1-6.py) via the non-wsq-courseware-build engine.	Dr. Alfred Ang

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Introduction

This Learner Guide accompanies the course Certified Lean Six Sigma Yellow Belt (CLSSYB) Training (C524), conducted by Tertiary Infotech Academy Pte Ltd. It follows the DMAIC roadmap end to end — Define, Measure, Analyze, Improve, Control — and provides step-by-step instructions for every hands-on lab.

The course content is grounded in the body of knowledge published by The Council for Six Sigma Certification (CSSC) in 'Six Sigma: A Complete Step-by-Step Guide', so what you learn here matches the recognised Yellow Belt standard. Every lab uses one continuous scenario — the Contoso Service Desk, where IT tickets take too long to be assigned and employees must chase for status. By the end of the course your lab outputs form a complete improvement package: role definition, VOC/CTQ, process maps, data collection plan, analysis, root cause, countermeasures and a control plan.

Course Learning Outcomes

- LO1: Define a project and establish the scope of work using Six Sigma's Define, Measure and Analyse phases.
- LO2: Apply Lean and Six Sigma concepts — value, waste, defects and variation — to a work process.
- LO3: Map a process using SIPOC, process maps and value stream maps to expose handoffs and waste.
- LO4: Collect and analyse process data using check sheets, Pareto charts, run charts and basic metrics.
- LO5: Identify root causes using 5 Whys, Fishbone analysis and evidence-based prioritisation.
- LO6: Recommend improvement and control actions to sustain gains, and prepare for certification.

Before You Start — Preparation

What you need

- A laptop or tablet with a spreadsheet application (Microsoft Excel, Google Sheets or LibreOffice Calc).
- A web browser for the interactive problem-solving tools listed in labs/tools.md.
- Printed or digital copies of this Learner Guide and the lab worksheets.
- Pen and paper (or a whiteboard) — several labs are faster sketched by hand.
- Optional: a real process from your own workplace you would like to improve.

Verify your setup

No software installation is required for this course. Before you begin, confirm you can open a blank spreadsheet and reach the browser-based tools listed in labs/tools.md.

Conventions used in every lab

- Every lab uses the same Contoso Service Desk scenario, so your outputs accumulate across the course.
- Each lab states what you will build; keep every output — later labs depend on earlier ones.
- Where a lab uses figures, the data is provided in the lab worksheet — no external data is needed.
- Use only process data you are authorised to share if you substitute your own workplace scenario.

Topic 01 — Six Sigma Foundations (15%)

Quality · Lean · Six Sigma · Lean Six Sigma · Belt roles · The DMAIC roadmap

Key concepts

- What is QualityQuality is conformance to customer requirements — not merely defect-free output.
- What is LeanA method to maximise customer value by systematically removing waste and improving flow.
- What is Six SigmaA data-driven method to reduce variation and defects — targeting 3.4 defects per million.
- Lean Six SigmaThe combined method: faster flow AND fewer defects, driven by data.
- Belt rolesWhite, Yellow, Green, Black and Master Black Belt — who does what on a project.
- The DMAIC roadmapDefine, Measure, Analyze, Improve, Control — the disciplined improvement path.

Lab 1 — Yellow Belt Role, Certification Paths, and Improvement Scenario

Learning outcome: Explain the Yellow Belt role and select a suitable improvement scenario (A1).

Goal: Establish what a Lean Six Sigma Yellow Belt is accountable for, how the belt pathway progresses from White to Master Black Belt, and choose a small, manageable improvement scenario to carry through the whole course.

What you'll build

A Yellow Belt responsibility table and a selected improvement scenario. (Tools: Belt role matrix, CSSC certification pathway, project selection criteria.)

Step-by-step

1. Define the Yellow Belt role — list at least four responsibilities and your contribution to each.
2. Compare the White, Yellow, Green, Black and Master Black Belt pathways in a table.
3. Review the Contoso Service Desk scenario: tickets are slow to be assigned and status is unclear.
4. Test your scenario against the 'good project' criteria — day-to-day work, manageable, aligned to business goals, data available.
5. Record why a Yellow Belt supports rather than leads this improvement.

Test it

You can state the Yellow Belt role in one sentence and justify your scenario against all four selection criteria.

Note: Full commands and screenshots are in labs/lab-01-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Topic 02 — Define — Scope the Problem (25%)

VOC · CTQ · Project charter · Problem statement · Scope · SIPOC · Process mapping

Key concepts

- Voice of the Customer Capture customer needs in their own words before deciding anything.
- Critical to Quality Translate each VOC need into a specific, measurable CTQ requirement.
- Project charter Problem statement, goal, scope, team and benefit — the project's contract.
- Problem statement Process, time period, measurable issue and impact — and never a solution.
- SIPOC The macro as-is map: Suppliers, Inputs, Process, Outputs, Customers.
- Process mapping Flowcharts and swimlanes expose the handoffs where delay and defects are born.

Lab 2 — Lean, Six Sigma, Waste, Voice of Customer, and Value

Learning outcome: Apply Lean and Six Sigma concepts — value, waste, defects, variation (K1, A2).

Goal: Separate what the customer actually values from the waste that surrounds it. Capture the Voice of the Customer, translate it into measurable CTQ requirements, and run a waste walk using the eight wastes (DOWNTIME).

What you'll build

A VOC-to-CTQ translation table, a value-added analysis, and a waste walk log. (Tools: VOC, CTQ tree, DOWNTIME eight wastes, value-added analysis.)

Step-by-step

1. Capture the Voice of the Customer — record at least five verbatim customer statements.
2. Translate each VOC statement into a need, then into a measurable CTQ requirement.
3. Classify each process activity as value-added, business-value-added or non-value-added.
4. Conduct a waste walk and tag each observation against the eight wastes (DOWNTIME).
5. Distinguish a defect (output fails CTQ) from waste (effort the customer will not pay for).
6. Identify which single waste type appears most often in your scenario.

Test it

Every CTQ is measurable with a target, and each waste observation is tagged to one of the eight waste types.

Note: Full commands and screenshots are in labs/lab-02-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Lab 3 — SIPOC, Process Mapping, Handoffs, and SME Support

Learning outcome: Map a process with SIPOC and a detailed process map to expose handoffs (A2, A3).

Goal: Build the macro 'as-is' view with SIPOC, then drill into a detailed process map that shows every actor, system and handoff — the points where delay and defects are actually created.

What you'll build

A completed SIPOC and a detailed process map with pain points marked. (Tools: SIPOC, process flowchart, swimlane map, standard process symbols.)

Step-by-step

1. Set the process boundaries — agree the explicit start and stop points first.
2. Build the SIPOC: Suppliers, Inputs, Process (5-7 high-level steps), Outputs, Customers.

3. Expand into a detailed process map: Step, Actor, Activity, System, Handoff.
4. Draw the same flow as a swimlane map so each actor's lane makes handoffs obvious.
5. Mark pain points — delays, rework loops, unclear ownership and handoff risks.
6. Prepare SME notes for the Green Belt: what you observed and what needs validation.

Test it

Your SIPOC has all five columns populated and every handoff on the detailed map has a named owner on both sides.

Note: Full commands and screenshots are in labs/lab-03-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Lab 4 — PDCA Small Improvement Project Charter

Learning outcome: Charter a small improvement using PDCA with a measurable goal (A1, A2).

Goal: Not every improvement needs a full DMAIC project. Use PDCA to charter a small, fast improvement that a Yellow Belt can run under guidance — with a problem statement, a measurable goal and a defined check point.

What you'll build

A one-page PDCA charter with problem statement, goal, scope and success measure. (Tools: PDCA cycle, problem/goal statements, scope in-out, stakeholder list.)

Step-by-step

1. Map the four PDCA phases to concrete actions for your scenario.
2. Write the problem statement: process, time period, measurable issue and impact — no solution.
3. Write a measurable goal statement: metric, baseline, target and date.
4. Define scope with an explicit in-scope / out-of-scope table to prevent scope creep.
5. Identify stakeholders and the decision you will make after the Check phase.
6. Confirm the improvement is small enough to test within two weeks.

Test it

Your goal statement contains a metric, a baseline, a target and a date, and your problem statement names no solution.

Note: Full commands and screenshots are in labs/lab-04-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Lab 5 — DMAIC Overview, Problem Statement, Scope, and Stakeholders

Learning outcome: Support the Define phase of a DMAIC project (A1, A2).

Goal: Step up from PDCA to the full DMAIC roadmap. Clarify what each phase delivers, where a Yellow Belt contributes most, and prepare defensible Define-phase information for the project leader.

What you'll build

A DMAIC phase table, refined problem statement, stakeholder map and benefit estimate. (Tools: DMAIC roadmap, project charter, stakeholder analysis, CTQ linkage.)

Step-by-step

1. Summarise all five DMAIC phases: purpose, key deliverable and Yellow Belt support role.
2. Refine the problem statement so it is specific, measurable and solution-free.
3. Identify stakeholders and classify each by influence and interest.
4. Define the project scope and state the expected benefit in business terms.
5. Link the problem back to the CTQ requirements captured in Lab 2.
6. Confirm which DMAIC phases a Yellow Belt can support most strongly.

Test it

Your problem statement passes the 'no solution named' test and every stakeholder has a defined engagement approach.

Note: Full commands and screenshots are in labs/lab-05-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Lab 11 — Affinity Diagram and Kano Analysis

Learning outcome: Organise customer requirements and classify them by satisfaction impact (K2).

Goal: Two Define-phase tools from the original course. Use an Affinity Diagram to cluster unstructured VOC input into themes, then apply Kano Analysis to classify requirements as Must-Be, One-Dimensional or Delighter.

What you'll build

An Affinity Diagram of clustered VOC themes and a Kano classification table. (Tools: Affinity Diagram, Kano Analysis, VOC clustering.)

Step-by-step

1. Write each VOC statement on a separate note — one idea per note.
2. Cluster related notes into natural groups without talking, then name each group.
3. Classify each requirement as Must-Be, One-Dimensional or Delighter.
4. Plot the requirements on the Kano diagram and identify where to invest first.

Test it

Every VOC note sits in exactly one named affinity group and carries a Kano classification.

Note: Full commands and screenshots are in labs/lab-11-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Topic 03 — Measure — Quantify Performance (25%)

The 8 wastes · Data types · Data collection plan · Check sheets · Yield · DPMO · Sigma level

Key concepts

- The 8 wastes — DOWNTIME Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.
- Types of data Continuous vs discrete; nominal vs ordinal — the data type drives the tool choice.
- Data collection plan What, who, when and how — with operational definitions to remove ambiguity.
- Check sheets A simple structured form is the most reliable way to capture process data.
- Process metrics Yield, DPU, DPO and DPMO quantify how well the process actually performs.
- Sigma level Convert DPMO into a sigma level to benchmark the process against Six Sigma.

Lab 6 — Data Collection, KPIs, Check Sheets, and Basic Metrics

Learning outcome: Plan and execute data collection against defined quality standards (A4).

Goal: Replace anecdote with evidence. Define KPIs with unambiguous operational definitions, design a check sheet, plan the sampling approach, and understand which data type you are collecting — because the data type drives the tool.

What you'll build

A data collection plan, an operational definition set and a working check sheet. (Tools: KPI definition, operational definitions, check sheets, sampling, data types.)

Step-by-step

1. Classify your data: continuous vs discrete, nominal vs ordinal — and why it matters.
2. Define KPIs with operational definitions, data source and collection frequency.
3. Design a check sheet capturing date, ticket ID, type, assignment time and defect category.
4. Define mutually exclusive defect categories so every observation lands in exactly one.
5. Plan sampling: how many, how often, by whom — and identify possible bias.
6. State which KPI best reflects the customer pain point from your VOC work.

Test it

Two different people reading your operational definition would record the same value for the same event.

Note: Full commands and screenshots are in labs/lab-06-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Lab 12 — Value Stream Map and Takt Time

Learning outcome: Quantify flow, lead time and takt time across the value stream (A3, A4).

Goal: Extend process mapping into Lean metrics. Map the value stream with process and wait times, calculate process cycle efficiency, then compute takt time to see whether the process can meet customer demand.

What you'll build

A value stream map with lead time, process time and a calculated takt time. (Tools: Value stream mapping, lead time, WIP, takt time, cycle efficiency.)

Step-by-step

1. Map the value stream: each step with its process time and the wait time between steps.
2. Total the value-added time and the lead time, then compute process cycle efficiency.
3. Calculate takt time = available working time / customer demand.
4. Compare cycle time against takt time to identify the bottleneck step.

Test it

Your lead time equals the sum of all process and wait times, and takt time is expressed per unit.

Note: Full commands and screenshots are in labs/lab-12-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Topic 04 — Analyze — Find the Root Cause (20%)

Variation · Pareto · Run charts · 5 Whys · Fishbone · Multi-voting · Evidence

Key concepts

- **Variation** Common cause is built into the process; special cause is an external, assignable signal.
- **Pareto analysis** The 80/20 rule separates the vital few causes from the trivial many.
- **Run charts** Plot the metric over time to reveal trend, shift, cluster and oscillation.
- **5 Whys** Ask why repeatedly to drill from the symptom down to an actionable cause.
- **Fishbone diagram** Organise candidate causes by category — Manpower, Method, Machine, Material, Measurement.
- **Evidence testing** A cause is only a root cause when the data you collected supports it.

Lab 7 — Pareto, Run Charts, Variation, Yield, DPU, and DPMO

Learning outcome: Analyse process performance data to quantify and prioritise (A3, A4).

Goal: Turn collected data into decisions. Apply the Pareto principle to find the vital few categories, use a run chart to see behaviour over time, and calculate the standard Six Sigma process metrics.

What you'll build

A Pareto chart, a run chart and calculated yield, DPU, DPO, DPMO and sigma level.
(Tools: Pareto Chart tool (collaborative).)

Step-by-step

1. Build the Pareto table: category, count, percentage and cumulative percentage.
2. In your group, open the collaborative Pareto tool — one member creates the session and shares the code, then everyone joins, brainstorms and votes to produce the live Pareto chart.
3. Identify the vital few categories driving about 80% of the problem.
4. Export your assignment-time data as CSV and plot a run chart in NovaSPC.

5. Interpret the run chart for trend, shift, cluster and oscillation patterns.
6. Calculate yield, DPU, DPO and DPMO from your defect data.
7. Convert DPMO to a sigma level and interpret what it says about the process.
8. Distinguish common-cause from special-cause variation and why the response differs.

Test it

Your cumulative percentage column reaches 100%, and you can state the sigma level with the DPMO it came from.

Note: Full commands and screenshots are in labs/lab-07-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Lab 8 — Root Cause Analysis with 5 Whys, Fishbone, and Evidence

Learning outcome: Identify root causes of variation using structured analysis (A3).

Goal: Move from symptom to cause. Apply the 5 Whys to drill past the obvious, organise candidate causes on a Fishbone diagram, then test each candidate against the evidence you actually collected.

What you'll build

A completed 5 Whys chain, a Fishbone diagram and an evidence-tested cause shortlist. (Tools: 5 Whys tool, Fishbone tool, 5M categories, brainstorming, multi-voting.)

Step-by-step

1. Write the problem statement precisely — the effect you are explaining.
2. Complete a 5 Whys chain with the online 5 Whys tool, continuing until you reach an actionable cause.
3. Build a Fishbone diagram with the online Fishbone tool using the 5M categories: Manpower, Method, Machine, Material, Measurement.
4. Brainstorm candidate causes into each category — no evaluation during generation.
5. Use multi-voting to shortlist the most likely causes as a team.
6. Test each shortlisted cause against your Lab 7 data — does the evidence support it?
7. State why the team must not jump straight to solutions.

Test it

Each shortlisted root cause is supported by named evidence, and your 5 Whys chain ends at something you can act on.

Note: Full commands and screenshots are in labs/lab-08-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Topic 05 — Improve — Fix the Cause (10%)

Countermeasures · 5S · Poka-Yoke · Standard work · Kaizen · Solution selection · Piloting

Key concepts

- Generating solutionsBrainstorm widely against the proven root cause before evaluating anything.
- Solution selectionScore candidate solutions against weighted criteria before committing.
- 5SSort, Set in order, Shine, Standardise, Sustain — for physical and digital work.
- Poka-YokeMistake proofing makes the error difficult or impossible to make in the first place.
- Standard workDocument the improved method so anyone can repeat it the same way.
- PilotingTest the change at small scale to expose issues before full rollout.

Lab 9 — Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen

Learning outcome: Recommend improvement actions that address the proven root cause (A5).

Goal: Generate and select countermeasures that attack the root cause you proved — not the symptom. Apply the core Lean improvement tools: 5S, poka-yoke, visual management, standard work and Kaizen.

What you'll build

A prioritised countermeasure set with a 5S plan, a poka-yoke design and standard work. (Tools: Brainstorming, 5S, poka-yoke, visual management, standard work, Kaizen.)

Step-by-step

1. Generate countermeasures for each proven root cause — quantity before quality.
2. Prioritise using an impact vs effort matrix to find the quick wins.
3. Apply 5S thinking — Sort, Set in order, Shine, Standardise, Sustain — including to digital work.
4. Design one poka-yoke that makes the error difficult or impossible to make.
5. Draft standard work so the improved method is repeatable by anyone.
6. Plan a Kaizen event or pilot to test the countermeasure at small scale first.

7. Distinguish a containment countermeasure from a permanent solution.

Test it

Every countermeasure traces back to a proven root cause, and your poka-yoke prevents rather than detects the error.

Note: Full commands and screenshots are in labs/lab-09-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Lab 13 — Solution Selection Matrix, Benchmarking, and FMEA

Learning outcome: Evaluate and de-risk candidate solutions before implementation (A5, K2).

Goal: Three Improve-phase tools from the original course. Score solutions against weighted criteria, benchmark against outstanding practice, and use FMEA to find how a proposed change could fail before it does.

What you'll build

A scored solution selection matrix, a benchmarking summary and an FMEA with RPN. (Tools: Solution selection matrix, benchmarking, FMEA, RPN scoring.)

Step-by-step

1. Agree the selection criteria and weight each by importance.
2. Score every candidate solution against the weighted criteria and rank the results.
3. Benchmark your process against an internal or external reference and note the gap.
4. Build an FMEA: failure mode, effect, cause, then score Severity, Occurrence and Detection.
5. Calculate $RPN = S \times O \times D$ and address the highest-RPN failure modes first.

Test it

Your matrix ranks solutions by weighted score, and every FMEA row has an RPN and an action for the highest scores.

Note: Full commands and screenshots are in labs/lab-13-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Topic 06 — Control — Hold the Gain (5%)

Control plan · Visual management · SOPs · Huddles · A3 · Handover · Certification

Key concepts

- Control plan Metric, target, monitoring method, frequency, owner and reaction plan.
- Process control A process is in control when it varies consistently within expected limits.
- Visual management Boards and dashboards keep the improved performance visible to everyone.
- Standard operating procedures Written instructions that lock in the improved method.
- Team huddles Short, regular stand-ups that surface problems while they are still small.
- A3 and handover One-page storytelling to summarise, hand over and sustain the improvement.

Lab 10 — Control Plan, A3 Summary, Handover, and Certification Readiness

Learning outcome: Recommend control actions to sustain process performance (A5).

Goal: Hold the gain. Build a control plan that names the metric, the target, the monitoring frequency, the owner and the reaction plan — then tell the whole improvement story on a single A3 page and hand it over.

What you'll build

A control plan, an A3 one-page summary, a handover checklist and a readiness plan. (Tools: Control plan, SPC thinking, visual management, A3 report, handover.)

Step-by-step

1. Create the control plan: metric, target, monitoring method, frequency, owner, reaction plan.
2. Define the visual management approach — board, dashboard or huddle — that keeps it visible.
3. Specify the reaction plan: exactly what happens when a metric falls outside target.
4. Write the A3 summary: background, current state, goal, analysis, countermeasures, results, follow-up.
5. Prepare the handover checklist so the process owner can sustain it without you.
6. Complete your personal certification readiness plan — which topics need most review.

Test it

Every control plan row has a named owner and a reaction plan, and your A3 fits on one page.

Note: Full commands and screenshots are in labs/lab-10-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Lab 14 — Descriptive Statistics and Implementation Planning

Learning outcome: Summarise data numerically and plan the rollout (A4, A5).

Goal: Close the loop. Compute the measures of central tendency and dispersion that describe your data, then convert the selected countermeasures into a dated implementation plan with owners and barriers identified.

What you'll build

A descriptive statistics summary and a dated implementation plan. (Tools: Mean, median, range, standard deviation, implementation planning.)

Step-by-step

1. Compute the mean and median for your assignment-time data and compare them.
2. Compute the range and standard deviation to describe the spread.
3. Explain what the mean-versus-median difference reveals about outliers and skew.
4. Write the implementation plan: action, owner, date, barriers and mitigation.

Test it

You can explain why the mean and median differ in your data, and every implementation action has an owner and a date.

Note: Full commands and screenshots are in labs/lab-14-*.md. Use only process data you are authorised to use if you substitute your own workplace scenario.

Wrap-Up — The DMAIC Roadmap and Sustaining the Gain

These cross-cutting themes run through every phase of the course. Study this section alongside the labs so the same approach transfers from the Contoso Service Desk scenario to a real improvement in your own workplace.

The DMAIC roadmap

- Define — establish the problem, the customer requirement (VOC to CTQ), the scope and the as-is process (SIPOC, process map).
- Measure — quantify today's performance with a data collection plan, check sheets, and metrics such as yield, DPU, DPO and DPMO.
- Analyze — separate common from special cause variation, then find the root cause with Pareto, run charts, 5 Whys and Fishbone.

- Improve — generate and select countermeasures, apply 5S, poka-yoke and standard work, and pilot before full rollout.
- Control — lock in the gain with a control plan, SOPs, visual management, team huddles and an A3 handover.

Habits that make improvement stick

- Measure before you change — without a baseline you cannot prove the improvement worked.
- Attack the process, not the people — the system produces the result you are seeing.
- Test every candidate cause against the evidence before you call it a root cause.
- Standardise the improved method, or the process will drift back within weeks.
- Make performance visible — what is seen daily is what gets sustained.

Next Steps

- First pass: complete every lab, following the steps in each lab worksheet.
- Second pass: repeat the DMAIC cycle on a small process from your own workplace.
- Build your improvement package — keep the VOC/CTQ, maps, analysis, countermeasures and control plan together.
- Progress towards Green Belt for deeper statistics, hypothesis testing and full project leadership.

Glossary

- **Lean** — A method to maximise customer value by systematically removing waste and improving flow.
- **Six Sigma** — A data-driven method to reduce variation and defects, targeting 3.4 defects per million opportunities.
- **DMAIC** — Define, Measure, Analyze, Improve, Control — the disciplined Lean Six Sigma improvement roadmap.
- **VOC** — Voice of the Customer — customer needs captured in the customer's own words.
- **CTQ** — Critical to Quality — a specific, measurable requirement translated from a VOC need.
- **SIPOC** — A macro as-is process map: Suppliers, Inputs, Process, Outputs, Customers.
- **DOWNTIME** — The eight wastes: Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.
- **Value-added** — An activity the customer would be willing to pay for, done right the first time.
- **Defect** — An output that fails to meet a CTQ requirement.

- **Variation** — The spread in process output; common cause is inherent, special cause is assignable.
- **Yield** — The proportion of units produced without defect.
- **DPU / DPO / DPMO** — Defects per unit, defects per opportunity, and defects per million opportunities.
- **Sigma level** — A benchmark of process capability derived from DPMO.
- **Pareto analysis** — The 80/20 rule — separating the vital few causes from the trivial many.
- **Run chart** — A plot of a metric over time, used to reveal trend, shift, cluster and oscillation.
- **5 Whys** — Asking why repeatedly to drill from a symptom down to an actionable root cause.
- **Fishbone diagram** — A cause-and-effect diagram organising candidate causes by category.
- **5S** — Sort, Set in order, Shine, Standardise, Sustain — workplace organisation for physical and digital work.
- **Poka-Yoke** — Mistake proofing — making an error difficult or impossible to make.
- **Standard work** — The documented best-known method so anyone can repeat it the same way.
- **Kaizen** — Continuous, incremental improvement driven by the people who do the work.
- **Control plan** — Metric, target, monitoring method, frequency, owner and reaction plan.
- **A3** — A one-page structured summary used to tell the improvement story and hand it over.
- **Takt time** — The rate of customer demand — available time divided by units required.
- **FMEA** — Failure Modes and Effects Analysis — scoring risk by severity, occurrence and detection.